Q= Diff b/w UDP & TCP.

TCP	UDP
1) Transmission Control Protocol	1) User Datagram Protocol or Universal Datagram Protocol
2) TCP is a connection-oriented protocol	2) UDP is a connectionless protocol.
3) As a message makes its way across the internet from one computer to another. This is connection based.	3) UDP is also a protocol used in message transfer or transport. This is not connection based which means that one program can send a load of packets to another and that would be the end of the relationship
4) TCP is suited for applications that require high reliability, and transmission time is relatively less critical.	4) UDP is suitable for applications that need fast, efficient transmission, such as games.
5) HTTP, HTTPs, FTP, SMTP, Telnet	5) DNS, DHCP, TFTP SNMP, RIP, VOIP
6) TCP rearranges data packets in the order specified.	6) UDP has no inherent order as all packets are independent of each other.
7) Speed for TCP is slower than UDP	7) UDP is faster because error recovery is not attempted.

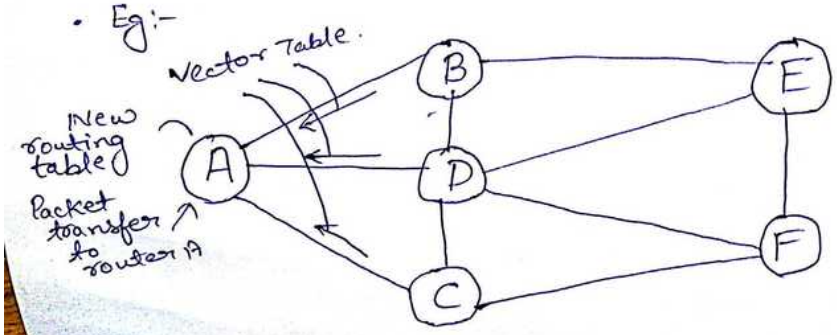
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Q:- Diff. b/w Leaky bucket and Token bucket Algorithm

Leaky bucket:	Token Bucket
Parameters: rate.	Parameters: rate, burstiness
1) Smooth out traffic by passing packets only when there is a token. Does not permit burstiness.	1. Token bucket smooths traffic too but permits burstiness - which is equivalent to the number of tokens accumulated in the bucket
2) Discards packets for which no tokens are available (no concept of queue).	2. Discard tokens when bucket is full, but never discard packets (infinite queue).
3) Application: Traffic shaping or Traffic policing.	3. Application: Network traffic shaping or rate limiting.

Q:- Describe Distance factor routing algorithm with suitable example.

- Distance Vector routing algorithm is a distributed algorithm.
- This routing algorithm is a iteration algorithm (when ever a packet comes to router the neighbouring router will give the vector table and a new vector table is created at that router.
- Eg:-



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- Vector table of B (A B C D E F
  1, 0, 3, 2, 4, 3).
- Vector table of D (2, 1, 2, 0, 1, 1).
- Vector table of C (3, 2, 0, 4, 5, 6).

measured delay of B, D, C from A are 3, 1, 2.

Now calculate the final routing table of A = ?

- Routing Table of A via B = (A B C D E F
  0, 3, 6, 5, 7, 6)
  - AB = AB + BB = 3 + 0 = 3
  - AC = AB + BC = 3 + 3 = 6
  - AD = AB + BD = 3 + 2 = 5
  - AE = AB + BE = 3 + 4 = 7
  - AF = AB + BF = 3 + 3 = 6 :
  - AA = 0;

- Routing Table of A via D : (A B C D E F
  0, 2, 3, 1, 2, 2)
  - AB = AD + DB = 1 + 1 = 2
  - AC = AD + DC = 1 + 2 = 3
  - AD = AD + DD = 1 + 0 = 1
  - AE = AD + DE = 1 + 1 = 2
  - AF = AD + DF = 1 + 1 = 2.

- Routing Table of A via C : (A B C D E F
  0, 4, 2, 6, 7, 8)
  - AB = AC + CB = 2 + 2 = 4
  - AC = AC + CC = 2 + 0 = 2
  - AD = AC + CD = 2 + 4 = 6
  - AE = AC + CE = 2 + 5 = 7
  - AF = AC + CF = 2 + 6 = 8.

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. Now the final routing table of A:
Taking minimum value of A via (B, C, D):
= (0, 2, 2, 1, 2, 2)
router used for the routing table of A
= (0, 2, 2, 1, 2, 2)
(—, D, C, D, D, D).

As similarly we find the routing table of all other router.

Q= Diff between fully qualified & partially qualified DNS

FQDNS	PQDNS.
1) A fully qualified domain name (FQDN) is the complete domain name for a specific computer, or host, on the internet. The FQDN consists of two parts: The hostname and the domain name.	1) If a label is not terminated by a null string, it is called partially qualified domain name (PQDN). A PQDN starts from a node, but it does not reach the root.
2) For example, an FQDN for a hypothetical mail server might be my mail.somecollege.edu. The host name is my mail, and host is located within the domain somecollege.edu.	2) It is used when the name to be resolved belongs to the same site as the client. Here the resolver can supply the missing part, called suffix, to create an FQDN.

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Q- Diff. between Supernetting and Subnetting.

Subnetting	Supernetting
1. The process of dividing the single network into the multiple subnets is called as subnetting.	1. The process of aggregating two or more networks for which it generates single IP address is called as supernetting.
2. It is difficult to maintain and administrate a big network with lots of systems connected. Dividing all these systems in a multiple networks make easy to maintain and administrate for the network administrator.	2. All the network in supernet must be using same IP address class. As it is an aggregating process it doesnot make sense to apply on the single network.
3. The no. of 1's in subnet mask is always greater than or equal to the no. of network identifier bits.	3. The number of 1's in supernet mask is always less than the number of network identifier bits.
4. For subnet mask, bits are borrowed from the host identifier bits	4. For supernet mask, bits are borrowed from the network identifier bits.
5. The subnet mask is applicable for a single network.	5. The supernet mask is appliable for two or more network. These all network should be of the same IP address class.

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(2) - Diff between Flow control and Congestion Control.

Basis for comparison	Flow Control	Congestion Control
Basic	It controls the traffic from a particular sender to a receiver	It controls the traffic entering the network.
Purpose.	It prevents the receiver from being overwhelmed by the data	It prevents the network from getting congested.
Responsibility	Flow control is the responsibility handled by data link layer and the transport layer.	Congestion control is the responsibility handled by network layer and transport layer.
Responsible	The sender is responsible for transmitting extra traffic at receivers side.	The transport layer is responsible transmitting extra traffic into the network.
Preventive measures	The sender transmits the data slowly to the receiver	Transport layer transmits the data into the network slowly.
Methods.	Feed back based flow control and Rate based flow control.	Provisioning, traffic aware routing and admission control.

3)

Write short notes on.

1) FTP:- File Transfer Protocol (FTP) is a standard Internet protocol for transmitting files between computers on the Internet over TCP/IP connections. FTP is a client-server protocol that relies on two communications chanel between client and server: a command channel for controlling the conversation and a data channel for transmitting file content.

2) SMTP:- SMTP is part of the application layer of the TCP/IP protocol. Using a process called "store and forward," SMTP moves your email on and across networks. It works closely with something called the Mail Transfer Agent (MTA) to send your communication to the right computer and email box. (Simple Mail Transfer Protocol)

3) HTTP;- The Hypertext Transfer Protocol (HTTP) is an application protocol for distributed, collaborative and hypermedia information systems. HTTP is the foundation of data communication for the World Wide Web. Hypertext is structured text that uses logical links between nodes containing text. HTTP is the protocol to exchange or transfer hypertext.

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